Date:

# Introduction:

Software quality processes ensure that software works correctly, performs efficiently, protects data, and remains secure.

In software industry, many of these processes are automated using testing tools, CI/CD pipelines, and AI-powered agents to detect problems quickly and reduce manual effort.

The four most critical software quality processes are:

- Functional Testing
- Performance Testing
- Data Quality Testing
- Security Testing

## Functional Testing

**Definition:** Functional testing checks whether software features work according to the specified

**Example:** In a university portal, functional testing checks whether

Date:_____

student can log in, register for courses, and view their results correctly.

| Testing activity | Automation tools / agents |
|---|---|
| Unit testing | JUnit, TestNG, pytest |
| Web UI testing | Selenium, Playwright, Cypress |
| API testing | Postman, REST Assured |
| AI-assisted test creation | GitHub Copilot & other AI coding assistants |

# Performance Testing

**Definition:** Performance testing checks how fast, stable, and responsive software remains under different workloads.

**Example:** A university portal is tested to determine whether it can handle 10,000 students logging in simultaneously without becoming slow or crashing.

Date:_____

| Testing activity | Automation tools/agents |
|---|---|
| Load and stress testing | Apache, JMeter, Grafana k6 |
| API performance testing | k6, JMeter |
| Browser performance testing | Lighthouse, k6 Browser |
| Performance monitoring | Grafana, New Relic |
| These tools support | automated load, tests, |
| repeated performance | checks, and monitoring. |

**What is automated?**

Simulating thousands of virtual users.
Measuring response time and throughput.
Finding bottlenecks under heavy traffic.
Checking whether performance meets defined limits.

# Data Quality Testing

**Definition:** Data quality testing checks whether data is accurate, complete, consistent, valid and free from unexpected duplicates.

**Example:** In a University database, testing verifies that every student has a valid student ID, required fields are not empty and student records

Date:_____

are not duplicated.

| Testing activity | Automation tools/agents |
|---|---|
| Data validation | Great Expectations |
| Database and data pipeline testing | dbt tests |
| Data quality monitoring | Soda |
| Database integrity checks | SQL scripts and constraints |

These tools automate data validation and can identify problems during data processing or pipeline execution.

## What is automated?

- Checking missing or duplicate values.
- Validating data types and formats.
- Verifying database relationships and constraints.
- Detecting unexpected changes in data.
- Checking data pipelines after new data arrives.

Date:

# Security Testing

**Definition:** Security testing identifies vulnerabilities and verifies that software protects its users, systems and data against unauthorized access and attacks.

**Example:** A university portal is tested to ensure that a student cannot access another student's marks by changing a URL or request.

| Testing activity | Automation tools / agents |
|---|---|
| Source code security analysis (SAST) | SonarQube, CodeQL |
| Web application security testing (DAST) | OWASP ZAP, Burp Suite |
| Dependency vulnerability scanning (SCA) | Dependabot, Snyk |
| Container and configuration scanning | Trivy |

Date:_____

Security tools can be integrated into CI/CD pipelines to detect common vulnerabilities before deployment.

## What is automated?

- Detecting insecure coding patterns.
- Finding known vulnerabilities in dependencies.
- Scanning web applications for common security weaknesses.
- Identifying insecure configurations.
- Alerting developers when security checks fail.